

Smart Sauna Temperature and Humidity Alarm

Project Overview

Project Title

Smart Sauna Temperature and Humidity Alarm

Project Significance

This system provides real-time monitoring of temperature and humidity in sauna rooms.

When environmental levels approach critical thresholds (the maximum limits of human tolerance), the system triggers warnings to alert management for timely intervention and adjustment.

This significantly mitigates the health risks associated with unsafe sauna environments, such as heatstroke and dehydration.

System Design

Hardware Components

- **U1 Sub-board (MCU):** Serves as the central microcontroller unit.
- **S2 Temperature & Humidity Sensor:** Collects real-time environmental data from the sauna room.
- **E1 LED Indicator:** Visually displays the current status (**Yellow: Normal conditions; Red: Temperature exceeds the safety threshold; Blue: Humidity exceeds the safety threshold**).
- **E4 Speaker/Buzzer:** Triggers audio alarms.

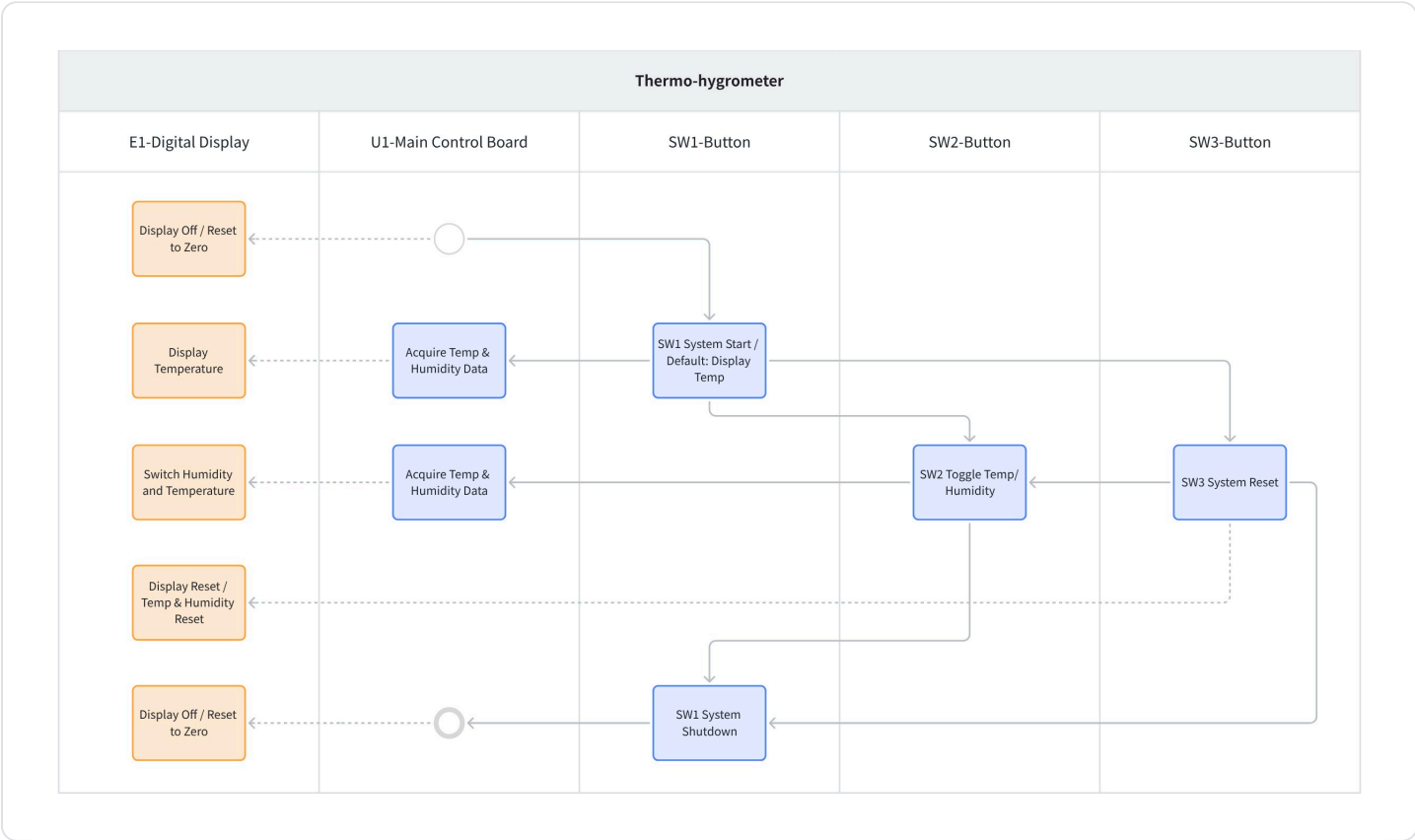
Working Principle

The U1 main control board drives the S2 sensor to continuously sample temperature and humidity data, and regulates the E1 LED to indicate the current environmental status based on the readings. If the collected data exceeds predefined safety thresholds, the MCU automatically activates the E4 speaker to sound an alarm.

Button Control Logic

Button	Function	Detailed Logic
SW1	Power Switch	Default state: Off. Reads the previous state. · If Off: Pressing turns the system On, initializes the program, and starts acquiring temperature/humidity data. · If On: Pressing turns the system Off and shuts down the program.
SW2	Display Toggle (Temp/ Humidity)	Reads the previous state and current data value. · If displaying Temperature: Pressing switches the display to Humidity. · If displaying Humidity: Pressing switches the display to Temperature. · If in Reset state: Re-acquires data and sets the digital display to show Temperature.
SW3	System Reset	· Pressing triggers a system reset, clearing both temperature and humidity data values to zero.

Thermo-hygrometer Workflow



Alarm Workflow

